Unit 1 Study Guide · Prepare for Thursday

Tuesday, November 3, 2026 · 55-minute workshop · Exam Thursday, November 5

Name: _____ Section: _____

> **Bring:** Monday's review, corrected problem sets 4–6, your cart-lab graphs, calculator, and paper for a reference-sheet draft. The exam covers Unit 1 kinematics. Honors students also prepare the extension tasks.

Workshop plan

Minutes	Task	Evidence
0–5	Retrieve graph meanings from memory	Write slope/area meanings with units
5–20	Finish review 5–6; honors begin H1–H2	Separate x/y KDEA with shared time
20–32	Compare and repair review solutions	Explain one sign error and one equation choice
32–45	Build and test a reference-sheet draft	Use it on the fresh checks below
45–52	Retry a weak skill without notes	Complete labeled KDEA independently
52–55	Exit ticket	One remaining question + Wednesday study plan

Can you do these without a worked solution?

Skill	Evidence to show	Ready / retry
Motion graphs	x–t slope → velocity; v–t slope → acceleration; signed v–t area → displacement; absolute areas → distance	_____
Constant acceleration	Select an equation by the missing variable; rearrange before substituting	_____
Free fall	State coordinate choice; distinguish speed from signed velocity; use nonzero acceleration at the peak	_____
Projectiles	Resolve components, solve vertical time, carry the same time into horizontal motion	_____
Lab interpretation	Read slope and intercept with units; connect evidence to constant-acceleration model	_____
Honors	Derive same-height flight time; interpret multiple roots; explain model restrictions	_____

Equation and meaning bank

For constant acceleration in one axis: $v_f = v_i + at$; $\Delta x = v_i t + \frac{1}{2}at^2$; $v_f^2 = v_i^2 + 2a\Delta x$; $\Delta x = \frac{1}{2}(v_i + v_f)t$. Use Δy instead of Δx for vertical motion.

For launch angle θ measured above horizontal: $v_{ix} = v_i \cos\theta$; $v_{iy} = v_i \sin\theta$. Under the ideal projectile model, $a_x = 0$ and $a_y = -9.8\ \text{m/s}^2$ with up positive. Axes share time, not velocity.

Fresh checks · solve before discussing

A. Stopping

A bicycle slows uniformly from 10.0 m/s to rest over 25.0 m. Find acceleration and stopping time.

B. Drop and launch

Two balls start 1.80 m above a level floor. One is dropped and the other launched horizontally at 3.0 m/s at the same instant. Find when each hits and the horizontal distance traveled by the launched ball. Explain the comparison.

Wednesday · Independent study

There is no regular Wednesday physics meeting in the published Block 2 schedule. Spend 20–30 minutes: retrieve equations and graph rules (5), redo your weakest problem without notes (10), compare and repair (5), then test your reference sheet on a fresh problem (5–10). Stop and record a specific question if you cannot explain a step.

Thursday: bring your approved reference sheet, calculator, pencil, and ruler. Confirm the allowed sheet size/sides and any section-specific directions with Mr. Avendano.

Source materials

- Problem Set 4 · Motion and Graphs
- Problem Set 5 · Uniform Acceleration
- Problem Set 6 · Free Fall and Projectiles
- Unit 1 review slides